Determinants of hyperparameter robustness in connectome reservoir computing

Miles Walter Churchland[1], Raul de Palma Aristides[2], Jordi Garcia-Ojalvo[1], Anna Ritz[3], Greg Anderson[4], Miguel C. Soriano[2]

[1]Universitat Pompeu Fabra [2]IFISC, CSIC–UIB [3]Biology, Reed College [4]Computer Science, Reed College

FULL PAPER

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MAIN FINDING

Higher performance accompanies greater sensitivity to hyperparameter choice.

The empirical *C. elegans* connectome occupies a relatively low-variance regime.

Research question

Which recurrent network features support robust computation?

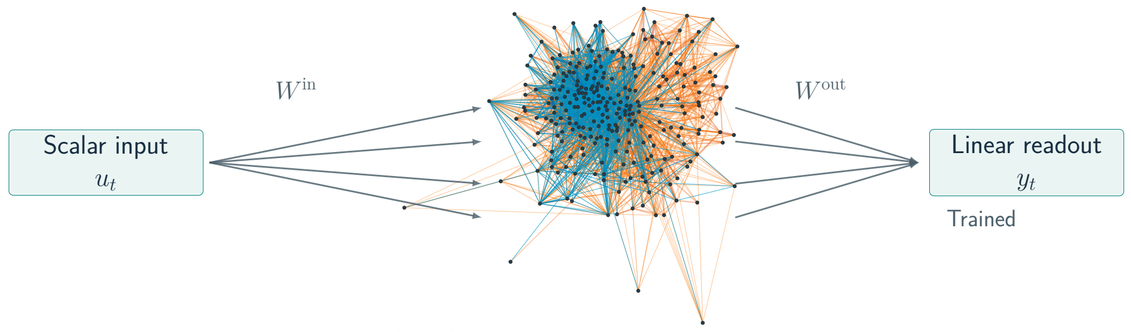
Reservoir performance can vary substantially with hyperparameters. We compare a structured *C. elegans* connectome with controlled network variants to see which features support stable computation.

96 settings × 200 trials

Vary target spectral radius, leak rate, input scale, and bias. Robustness is the coefficient of variation (CV) across the 96 settings; **lower CV means greater robustness**.

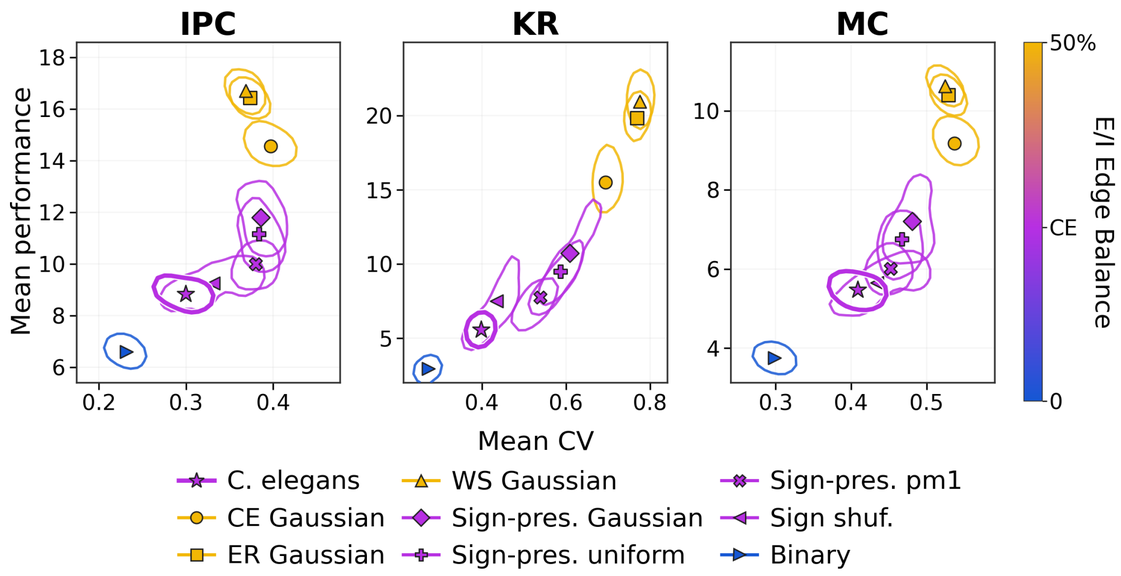
IPC: information processing capacity; nonlinear recall [2].
KR: kernel rank; history separation [3].
MC: memory capacity; recall of the past 30 inputs.

Reservoir computing



Input reaches **all 297 neurons** via Gaussian weights of either sign, fixed per trial up to input scaling; the trained readout combines their states. Gray arrows show direction, not sign. *C. elegans* connectome: 3,604 chemical edges [1]; largest component shown. Recurrent edges: positive; negative. Edge directions omitted.

Architecture tradeoff



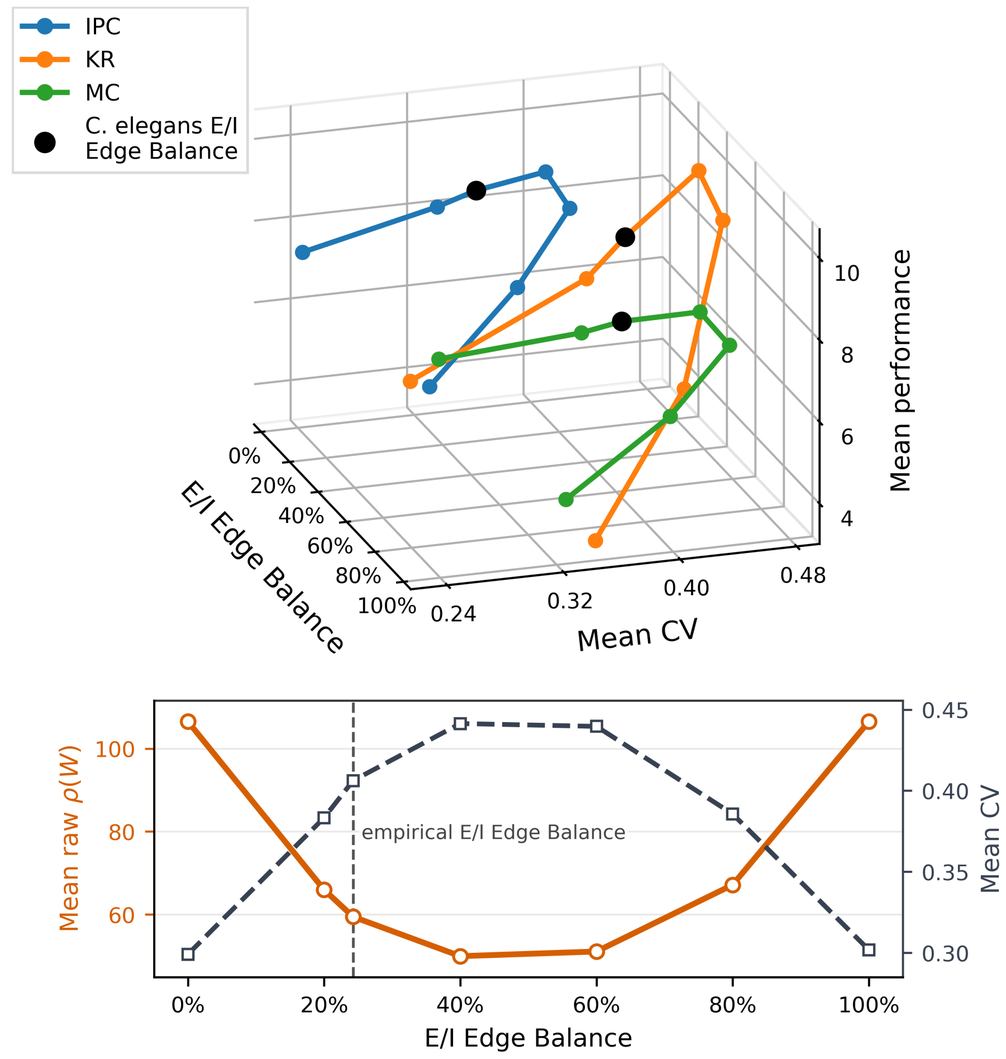
Higher performance accompanies higher CV. Mean performance is averaged across 96 settings. Contours enclose the highest-density 50% of trials; markers identify architectures. Colors show excitatory/inhibitory (E/I) edge balance: 0%, CE $\approx 24.25\%$, and 50%. CE: *C. elegans*; ER: Erdős–Rényi; WS: Watts–Strogatz. Sign-pres.: signs preserved; sign shuf.: signs shuffled; pm1: ± 1 weights.

Takeaway. The empirical connectome lies on the lower-sensitivity side of the performance–robustness tradeoff.

How to read the tradeoff

In the architecture plots, moving up means higher performance; moving right means greater sensitivity to hyperparameter settings. The empirical connectome lies toward the lower-left.

E/I edge balance



Changing E/I edge balance moves performance and CV together. Top: IPC, KR and MC; black markers indicate the empirical balance. Bottom: raw spectral radius and CV averaged across metrics. Points average 200 trials.

Normalization matters. At the same target radius, lower raw radius requires a larger recurrent-weight scaling factor. The observed tradeoff combines sign-structure changes and normalization-induced rescaling; these effects are not isolated.

Scope and limitations. Robustness here means low relative variation across the tested hyperparameter grid; it does not imply invariance to all parameter changes. IPC omits products across different delays. Of 3,604 chemical edges, 1,864 have unknown or complex signs and are assigned signs to match the labeled-edge ratio; selected controls remove them. Electrical synapses and self-loops are excluded; input reaches all units.

Selected references [1] Fenyves et al. (2020). Synaptic polarity and sign-balance prediction. *PLOS Comput. Biol.* 16, e1007974.
[2] Dambre et al. (2012). Information processing capacity of dynamical systems. *Sci. Rep.* 2, 514.
[3] Roy & Vetterli (2007). The effective rank: a measure of effective dimensionality. *EUSIPCO*, 606–610.